

aTableau

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Version 2.0-alpha



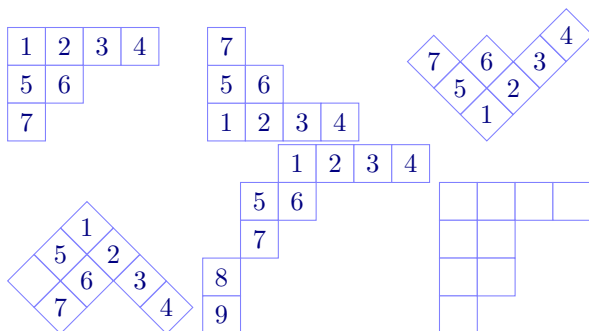
This version of the code is already quite powerful, but it is still in the proof-of-concept phase. There are still many bugs, planned features that are not yet implemented, and everything is subject to change.

CONTENTS

1. Tableaux	1
2. Young diagrams	2
3. Skew tableau	3
4. Shifted tableau	3
5. Tabloids	3
6. Ribbon tableaux	3
7. Multidiagrams and multitableaux	3
8. Abacuses	4
9. Dynkin quivers	4
10. Braid diagrams	4
11. KLRW diagrams	4
12. The code	4

Young diagrams, tableaux and abacuses are common objects used in algebraic combinatorics and representation theory. Partly because of their ubiquity, there are several different conventions in use and authors often want to decorate these objects in weird and wonderful ways.

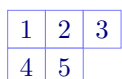
This package provides highly customisable ways of drawing combinatorial objects. The commands are designed to be easy to use, read and understand. Frequently it is necessary to further embellish these diagrams, so these commands either be used as standalone commands in the document or, by providing coordinates, they can be used inside `tikzpicture` environments when more advanced formatting is required.



```
\Tableau{1234,56,7}
\Tableau[french]{1234,56,7}
\Tableau[ukraine]{1234,56,7}
\Tableau[australian]{1234,56,.7}
\Tableau[skew={2,1^2}]{1234,56,7,8,9}
\Tableau{.....}
```

There are several other L^AT_EX packages available for drawing tableaux, Young diagrams etc.

1. TABLEAUX



```
\Tableau{123,45}
```

		10	11	
		1	9	
		8		
1	3	2,3	4	1001
5				
7				

								x	x	x
								x	x	x
								x	x	

1	2	3	1	2	3
4	5		4	5	

	5	3
4	2	
	1	

	1	
4	2	
	5	3

4	5
1	2

```
\begin{tikzpicture}
  \node at (1,1) {$\bullet$};
  \node at (2,2.5) {$\bullet$};
  \Tableau(1,1) {13{2,3}4{1001},5,7}
  \Tableau(2,2.5) {{10}{11},19,8}
\end{tikzpicture}
```

```
\Tableau[starstyle={fill=blue}]{... , ... , ...}
\Tableau[dotstyle={fill=blue!20}]{...},
.[atableau/dotstyle]\cdot ., ..}
\Tableau[dot=x]{... , ... , ...}
```

```
\tikzset{R/.style={fill=blue!20,draw=red,thick}}
\Tableau{12[R]3,4[R]5}
\Tableau[starstyle={fill=blue!50,text=white}]{12*3,4*5}
```

```
\tikzset{R/.style={fill=red!20,draw=red}}
\Tableau[ukraine]{12[R]3,45}
```

```
\tikzset{R/.style={fill=blue!20,draw=NavyBlue, thick}}
\Tableau[Australian]{12[{R, thick}]3,45}
```

```
\tikzset{R/.style={fill=red!20,circle, radius=0.2}}
\Tableau[french]{12[R]3,45}
```

2. YOUNG DIAGRAMS

x	x	x	x
x	x	x	
x	x	x	
x	x	x	
x	x	x	
x	x	x	
x			
x			
x			
x			

	x	
x	x	x
x	x	x

x	x	x
x	x	
x		

x		
x	x	
x	x	x

x	x	x
x	x	x
x	x	

```
\Diagram{4,3^5,1^4}
\Diagram{4,3,3,1}
```

```
\Diagram[australian]{3,2,1}
```

```
\Diagram[english]{3,2,1}
```

```
\Diagram[french]{3,2,1}
```

```
\Diagram[ukraine]{3,2,1}
```

0	1	2
-1	0	
-2		

```
\Diagram[label=content]{3,2,1}
```

5	4	1
3	2	
2	1	

```
\Diagram[label=hook]{3,2,2}
```

0	1	2
2	0	
1		

```
\Diagram[label=residue, char=3]{3,2,1}
```

3. SKEW TABLEAU

		1	2	3		6	7			
	4	5					4	5		
6	7							1	2	3

```
\tikzset{R/.style={fill=red!20,circle}}
\SkewTableau{2,1}{12[R]3,45,67}
\SkewTableau[french]{2,1}{12[R]3,45,67}
```

4. SHIFTED TABLEAU

1	2	3
	4	5

```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau{12[R]3,45}
```

	4	5
1	2	3

```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau[french]{12[R]3,45}
```

5. TABLOIDS

6. RIBBON TABLEAUX

7. MULTIDIAGRAMS AND MULTITABLEAUX

<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>								<table><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>						<table><tr><td></td></tr></table>			...		<table><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>				
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```
\Multidiagram[scale=0.5]{3,2^2|2,1^2|1}
```

```
\Multidiagram[scale=0.5,rows=3,rows=3]{3,2^2|\dots|2,1^2}
```

```
\Multidiagram[ukraine,scale=0.5,rows=3,
cols=3]{3,2^2|\dots|2,1^2}
```

```
\Multidiagram[french,scale=0.7,rows=3]{3,2^2|\dots|2,1^2}
```

1	2	3
4	5	
6	7	

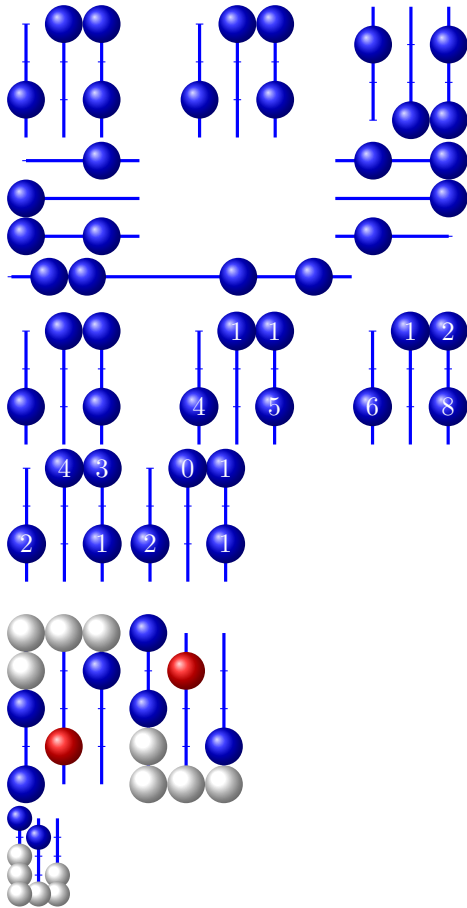
8	9
1	0
1	1

1	2
---	---

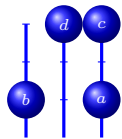
```
\Multitableau[rows=3,scale=0.5,font=\tiny]
{123,45,67|89,{10},{11}|\dots|{12}}
```

1		1		1		1
---	--	---	--	---	--	---

8. ABACUSES



An abacus:



Done!

```
\Abacus{3}{5,4,1,1}
\Abacus[style=down]{3}{5,4,1,1}
\Abacus[style=up]{3}{5,4,1,1}
\Abacus[style=right]{3}{5,4,1,1}
\Abacus[style=left]{3}{5,4,1,1}
\Abacus[style=right]{1}{5,4,1,1}
```

```
\Abacus{3}{5,4,1,1}
\Abacus[label=shape,text=white]{3}{5,4,1,1}
\Abacus[label=beta,text=white]{3}{5,4,1,1}
\Abacus[label=row,text=white]{3}{5,4,1,1}
\Abacus[label=residue,text=white]{3}{5,4,1,1}
```

```
\tikzset{R/.style={ball color=red},
S/.style={ball color=white}}
```

```
\Abacus{3}{5,[R]4,1^2,[S]0^4}
\Abacus[style=up]{3}{5,[R]4,1^2,[S]0^4}
```

```
\Abacus[style=up, scale=0.5, starstyle={ball
color=white}]{3}{5,4,*1^2,*0^4}
```

```
\Abacus[text=white,font=\tiny]{3}{5_a,4_b,1_c,1_d}
```

9. DYNKIN QUIVERS

10. BRAID DIAGRAMS

11. KLRW DIAGRAMS

12. THE CODE

Every command provided by `aTableau` uses `TikZ`, with the bulk of the code in the package being written in $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}3$. Most of the tableau commands are special cases of the `\Tableau` command. Here is a dictionary:

Command	Equivalent <code>\Tableau</code> command
<code>\ShiftedTableau{...}</code>	<code>\Tableau[shifted]{...}</code>
<code>\SkewTableau{μ}{...}</code>	<code>\Tableau[skew=μ]{...}</code>
<code>\Tabloid{...}</code>	<code>\Tableau[tabloid]{...}</code>

The `\Diagram` command also invokes the `\Tableau` command, however, this requires some involved internal trickery. For example, `\Diagram{3,2^2}` is the same as `\Tableau{..., ..., ...}`.